



Manifold Optimization on the Magnitude Torus for Fourier Phase Retrieval

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1. Fourier Phase Retrieval

The Fourier phase retrieval problem aims to find a vectorized, real D -dim. signal $\mathbf{x} \in \mathbb{R}^{n \times 1}$ subject to

$$\mathbf{y} = |\mathbf{A}\mathbf{x}| \text{ and } \mathbf{x} \in \mathcal{B},$$

where

- $\mathbf{y} \in \mathbb{R}_{>0}^{n \times 1}$: Fourier magnitude;
- $\mathbf{A}$: the normalized D -dim. DFT matrix;
- $\mathcal{B} \subseteq \mathbb{R}^{n \times 1}$: the signal domain constraint set. Some common examples include $\mathcal{B}_+ = \mathbb{R}_{\geq 0}^{n \times 1}$ and $\mathcal{B}_{01} = \{0, 1\}^{n \times 1}$.

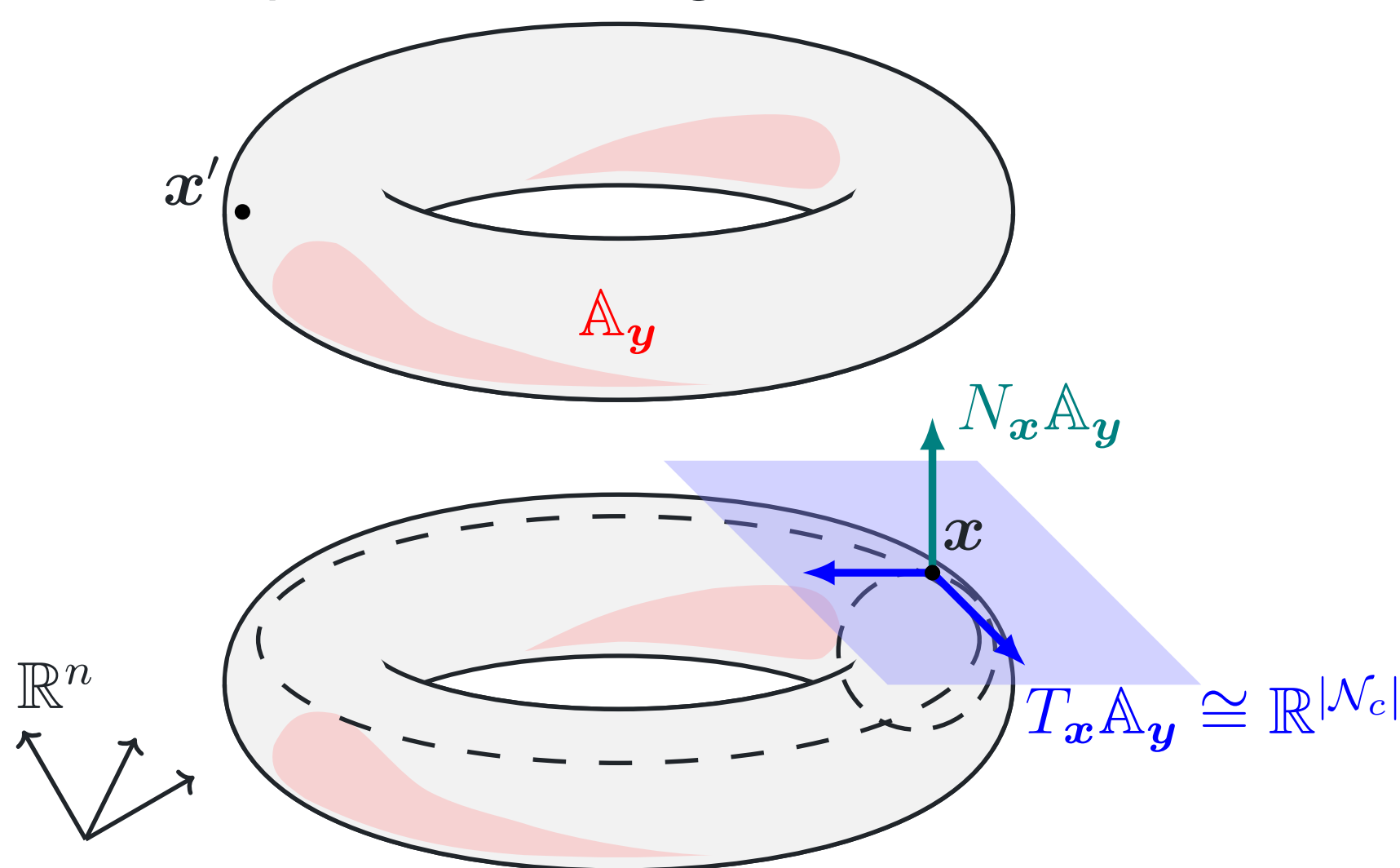
The Fourier domain constraint set is

$$\mathbb{A}_{\mathbf{y}} := \left\{ \mathbf{x} \in \mathbb{R}^{n \times 1} \mid |\mathbf{A}\mathbf{x}| = \mathbf{y} \right\},$$

termed the **magnitude torus**.

2. Geometry of Magnitude Torus

Understanding the **geometry of the intersection** between $\mathbb{A}_{\mathbf{y}}$ and $\mathcal{B}$ is crucial for determining the behavior of optimization algorithms.



Barnett et al. [1] described the tangent and normal spaces to $\mathbb{A}_{\mathbf{y}}$ at $\mathbf{x}$ as:

$$T_{\mathbf{x}} \mathbb{A}_{\mathbf{y}} = \text{span} \left\{ \mathbf{x}^{(v)} - \mathbf{x}^{(-v)} \right\}_v,$$

$$N_{\mathbf{x}} \mathbb{A}_{\mathbf{y}} = \text{span} \left\{ \mathbf{x}^{(v)} + \mathbf{x}^{(-v)} \right\}_v,$$

where $\mathbf{x}^{(v)}$ denotes $\mathbf{x}$ shifted by $\mathbf{v}$.

We extend these results. If the signal is of size $n = N_1 \times \dots \times N_D$, we can separate the rows of $\mathbf{A}$, denoted by $\mathbf{a}_k^H$, into two sets:

$$\mathbf{a}_j = \text{Re}\{\mathbf{a}_j\} = \mathbf{u}_j, \quad j \in \mathcal{N}_p;$$

$$\mathbf{a}_k = \text{Re}\{\mathbf{a}_k\} + i\text{Im}\{\mathbf{a}_k\} = \mathbf{u}_k + i\mathbf{v}_k, \quad k \in \mathcal{N}_c.$$

The magnitude torus is thus a union of $2^{|\mathcal{N}_p|}$ tori, each with dimension $|\mathcal{N}_c|$.

All points on the magnitude torus can be parameterized by $(\mathbf{s}, \mathbf{t}) \in \{1, -1\}^{|\mathcal{N}_p|} \times \mathbb{T}^{|\mathcal{N}_c|}$:

$$\mathbf{x}(\mathbf{s}, \mathbf{t}) = \sum_{j \in \mathcal{N}_p} s_j y_j \mathbf{u}_j + \sum_{k \in \mathcal{N}_c} 2y_k (\cos(t_k) \mathbf{u}_k + \sin(t_k) \mathbf{v}_k).$$

Note that the vectors $\{\mathbf{u}_k, \mathbf{v}_k\}_{k \in \mathcal{N}_p \cup \mathcal{N}_c}$ are mutually orthogonal, allowing for the clean results below:

The tangent and normal spaces have the following characterizations:

$$T_{\mathbf{x}} \mathbb{A}_{\mathbf{y}} = \ker \left(\text{Re}\{\text{diag}(\mathbf{A}\mathbf{x})\mathbf{A}^*\} \right),$$

$$N_{\mathbf{x}} \mathbb{A}_{\mathbf{y}} = \ker \left(\text{Im}\{\text{diag}(\mathbf{A}\mathbf{x})\mathbf{A}^*\} \right).$$

3. Manifold Optimization Algorithms

The retrieval problem is equivalent to finding the intersection $\mathbb{A}_{\mathbf{y}} \cap \mathcal{B}$. Many algorithms have been proposed to tackle the Fourier phase retrieval problem. Let $P_{\mathcal{A}}$ and $P_{\mathcal{B}}$ denote the orthogonal projections onto $\mathbb{A}_{\mathbf{y}}$ and $\mathcal{B}$, respectively.

- Gerchberg–Saxton: $\mathbf{x}_{k+1} = P_{\mathcal{B}} \circ P_{\mathcal{A}}(\mathbf{x}_k) = P_{\mathcal{B}} \left(\mathbf{A}^{-1} \left(\mathbf{y} \circ \text{sgn}(\mathbf{A}\mathbf{x}_k) \right) \right)$;
- Projected Gradient Descent (PGD): $\mathbf{x}_{k+1} = P_{\mathcal{B}} \left(\mathbf{x}_k - \mu \nabla \ell(\mathbf{x}_k) \right)$ with $\ell(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{y}\|^2$;
- Continuous Hybrid Input-Output (CHIO):

$$\mathbf{x}_{k+1} = \begin{cases} P_{\mathcal{A}}(\mathbf{x}_k), & \alpha \mathbf{x}_k \leq P_{\mathcal{A}}(\mathbf{x}_k); \\ \mathbf{x}_k - \left(\frac{1-\alpha}{\alpha} \right) P_{\mathcal{A}}(\mathbf{x}_k), & 0 \leq P_{\mathcal{A}}(\mathbf{x}_k) \leq \alpha \mathbf{x}_k; \\ \mathbf{x}_k - \beta P_{\mathcal{A}}(\mathbf{x}_k), & \text{otherwise.} \end{cases}$$

From the torus parameterization, we develop the following two algorithms: given a function $f: \mathbb{A}_{\mathbf{y}} \rightarrow \mathbb{R}$ that measures the mismatch between a point $\mathbf{x}$ and $\mathcal{B}$ and its extension F to a neighborhood of $\mathbb{A}_{\mathbf{y}}$, we have

$$\mathbf{t}_{k+1} = (\mathbf{t}_k - \mu \cdot \nabla (f \circ \mathbf{x})(\mathbf{t}_k)) \bmod 2\pi \quad \text{TorusGD}$$

$$\mathbf{x}_{k+1} = P_{\mathcal{A}} \left(\mathbf{x}_k - \mu \cdot \nabla F(\mathbf{x}_k) \right) \quad \text{dualPGD}$$

dualPGD is a Riemannian gradient descent algorithm since the map $P_{\mathcal{A}}$ is a **retraction**, i.e., for $\mathbf{v} \in T_{\mathbf{x}} \mathbb{A}_{\mathbf{y}}$,

$$(P_{\mathcal{A}})_{*,\mathbf{x}}(\mathbf{v}) := \left. \frac{d}{dt} \right|_{t=0} P_{\mathcal{A}}(\mathbf{x} + t\mathbf{v}) = \text{proj}_{T_{\mathbf{x}} \mathbb{A}_{\mathbf{y}}}(\mathbf{v}) = \mathbf{v}.$$

Some properties:

1. They essentially solve the **dual problem** to PGD: $\min_{\mathbf{x} \in \mathbb{A}_{\mathbf{y}}} f(\mathbf{x})$.
2. Both algorithms are Riemannian gradient descent algorithms under different metrics.
3. dualPGD is a scalable algorithm.
4. dualPGD is a **Douglas–Rachford-type** algorithm: let $\mathbf{x}_{k+1/2} := \mathbf{x}_k - \mu \nabla F(\mathbf{x}_k)$, then

$$\mathbf{x}_{k+1/2} = ((1 - \mu) \cdot P_{\mathcal{A}} + \mu \cdot P_{\mathcal{B}} \circ P_{\mathcal{A}})(\mathbf{x}_{k-1/2}), \quad \nabla F(\mathbf{x}) = \mathbf{x} - P_{\mathcal{B}}(\mathbf{x}).$$

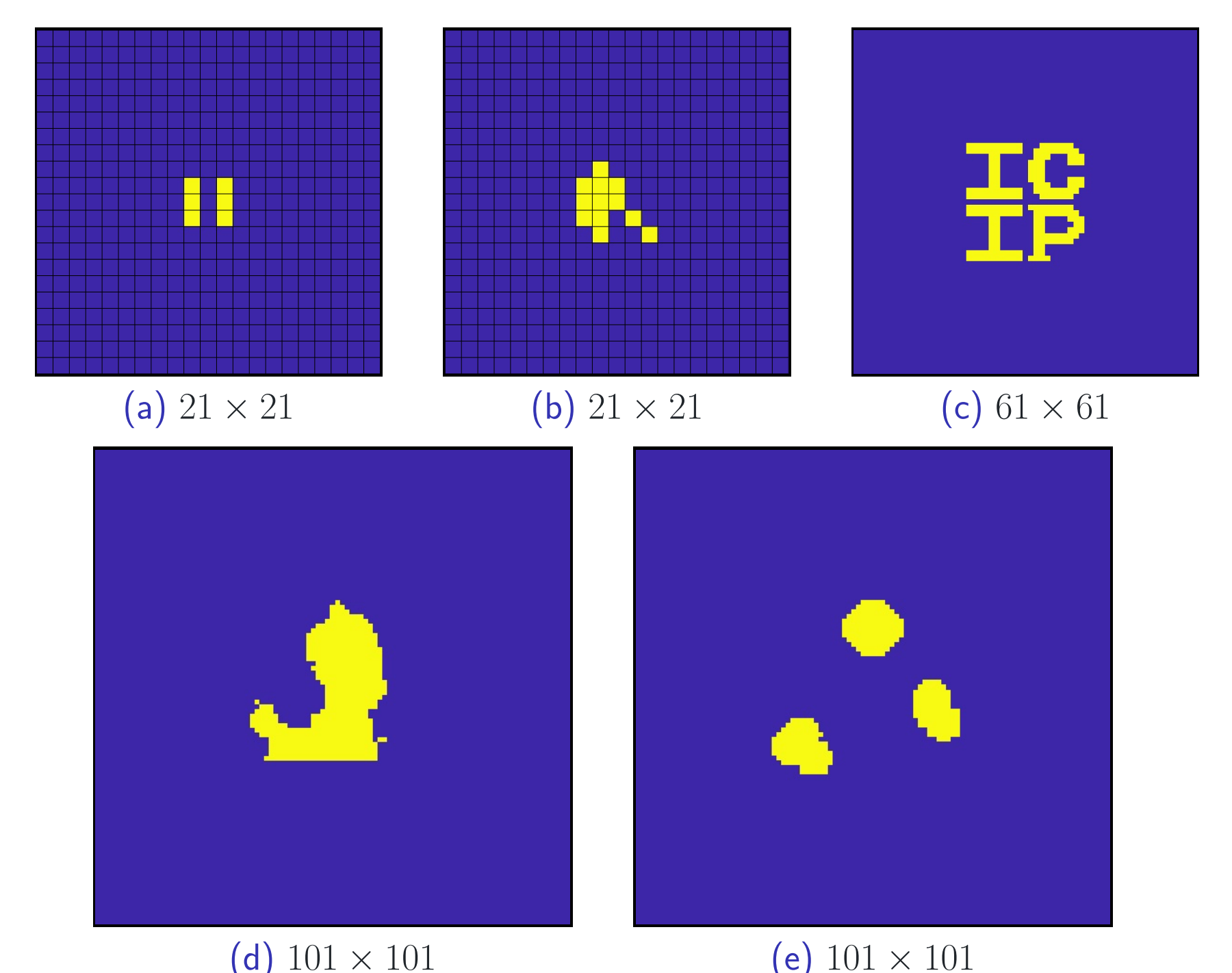
4. Simulation Results

Given binary signals with uniqueness guaranteed [2], we evaluate the retrieval rates over 1000 trials.

The initial conditions are sampled via

$$\mathbf{x}_0 \sim P_{\mathcal{A}} \left(\text{Unif}([0, 1]^{n \times 1}) \right).$$

Algorithm	Retrieval Rate (%)				
	(a)	(b)	(c)	(d)	(e)
TorusGD	100.0	98.5	-	-	-
dualPGD	100.0	99.2	8.0	84.5	99.8
PGD	85.2	95.0	1.4	74.5	92.4
GS	86.8	96.6	7.0	91.0	98.9
CHIO	100.0	100.0	72.7	100.0	100.0



5. Supplementary

Code and supplementary material are available at: https://wenperng.github.io/assets/pdf/papers/2026_ICIP_supp.pdf

References

- [1] A. H. Barnett et al. "Geometry of the phase retrieval problem". In: *Inverse Problems* 36.9 (Sept. 2020), p. 094003.
- [2] M. Hayes. "The reconstruction of a multidimensional sequence from the phase or magnitude of its Fourier transform". In: *IEEE Transactions on Acoustics, Speech, and Signal Processing* 30.2 (1982), pp. 140–154.